26 Q-Learning in 2-Player Games

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Tabular Q-learning in matrix games: convergence in coordination games, divergence in zero-sum games, and the rational learning debate.

Learning objectives

- Explain how single-agent Q-learning extends to multi-agent matrix games.
- Implement independent Q-learners for repeated 2-player games in R.
- Demonstrate convergence to Nash equilibrium in coordination games and cycling in zero-sum games.
- Discuss the “rational learning” debate and the limits of independent learning.

26.1 Motivation

Reinforcement learning has achieved remarkable success in single-agent settings: an agent interacts with an environment, receives rewards, and learns a policy that maximizes long-term return. But what happens when the “environment” is another learning agent?

In a matrix game played repeatedly, two Q-learners each try to maximize their own payoff. The environment each agent faces is **non-stationary** — it changes as the other agent updates its policy. This creates a fundamental tension: the convergence guarantees of Q-learning (25) rely on a stationary environment, and that assumption breaks in multi-agent settings.

Whether independent Q-learners converge to Nash equilibrium, cycle, or diverge depends on the game’s structure. This chapter explores all three outcomes through simulation.

26.2 Theory

26.2.1 Q-learning recap

In single-agent Q-learning (Sutton & Barto, 2018, Chapter 6), an agent maintains a value estimate $Q(s, a)$ for each state-action pair and updates it after observing a reward r and next state s':

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right] \tag{25.4}$$

For repeated matrix games, there is only one “state” (the game itself), so Q-values reduce to action values $Q(a_i)$ for each of player i’s actions. The discount factor γ is irrelevant in a stateless setting, and the update simplifies to:

$$Q_i(a_i) \leftarrow Q_i(a_i) + \alpha \left[r_i - Q_i(a_i) \right] \tag{26.1}$$

where r_i is the payoff player i received from playing action a_i against the opponent's (unknown) action.

26.2.2 Independent learners

The simplest multi-agent approach is **independent Q-learning**: each agent runs its own Q-learning algorithm, treating the other agent as part of the environment. No agent models or observes the other's Q-values.

The action selection typically uses an ε-greedy policy: with probability ε the agent explores (chooses a random action), and with probability $1 - \varepsilon$ it exploits (chooses the action with the highest Q-value).

26.2.3 Convergence properties

The behavior of independent Q-learners depends critically on the game:

- **Coordination games** (e.g., Battle of the Sexes): learners tend to converge to one of the pure-strategy Nash equilibria. Which one depends on initial conditions and exploration noise.
- **Zero-sum games** (e.g., Matching Pennies): learners typically **cycle** rather than converge. Each agent's best response shifts the opponent's optimal play, creating a feedback loop.
- **The Prisoner's Dilemma**: learners converge to mutual defection — the unique Nash equilibrium — even though mutual cooperation yields higher payoffs.

Shoham and Leyton-Brown (2009) discuss the "rational learning" debate: under what conditions can independent learners reach equilibrium, and when should we expect them to fail?

26.3 Implementation in R

26.3.1 Q-learning agent

```
q_agent_create ← function(n_actions, alpha = 0.1, epsilon = 0.1) {
  Q ← rep(0, n_actions)
  list(
    choose = function() {
      if (runif(1) < epsilon) {
        sample(n_actions, 1)
      } else {
        ties ← which(Q == max(Q))
        sample(ties, 1)
      }
    },
    update = function(action, reward) {
      Q[action] ⟵ Q[action] + alpha * (reward - Q[action])
    },
    get_Q = function() Q
  )
}
```

26.3.2 Running a repeated game

```
run_q_learning_game ← function(A, B, episodes = 5000,
                               alpha = 0.1, epsilon = 0.1) {
  n1 ← nrow(A)
  n2 ← ncol(A)
  agent1 ← q_agent_create(n1, alpha, epsilon)
  agent2 ← q_agent_create(n2, alpha, epsilon)

  history ← tibble(
    episode = integer(),
    a1 = integer(), a2 = integer(),
    r1 = double(), r2 = double(),
    Q1_1 = double(), Q1_2 = double(),
    Q2_1 = double(), Q2_2 = double()
  )

  for (ep in seq_len(episodes)) {
    a1 ← agent1$choose()
    a2 ← agent2$choose()
    r1 ← A[a1, a2]
    r2 ← B[a1, a2]
    agent1$update(a1, r1)
    agent2$update(a2, r2)

    Q1 ← agent1$get_Q()
    Q2 ← agent2$get_Q()
    history ← bind_rows(history, tibble(
      episode = ep, a1 = a1, a2 = a2, r1 = r1, r2 = r2,
      Q1_1 = Q1[1], Q1_2 = Q1[2],
      Q2_1 = Q2[1], Q2_2 = Q2[2]
    ))
  }
  history
}
```

26.3.3 Coordination game: convergence

```
A_coord ← matrix(c(2, 0, 0, 1), nrow = 2, byrow = TRUE,
                 dimnames = list(c("A", "B"), c("A", "B")))
B_coord ← A_coord

set.seed(42)
hist_coord ← run_q_learning_game(A_coord, B_coord,
                                 episodes = 3000, alpha = 0.1, epsilon = 0.05)
```

```
q_long ← hist_coord ▷
  select(episode, Q1_1, Q1_2, Q2_1, Q2_2) ▷
  pivot_longer(-episode, names_to = "variable", values_to = "Q") ▷
  mutate(
    player = if_else(str_starts(variable, "Q1"), "Player 1", "Player 2"),
    action = if_else(str_ends(variable, "_1"), "Action A", "Action B")
  )

p_conv ← ggplot(q_long, aes(x = episode, y = Q, colour = action, linetype = player))
+
  geom_line(alpha = 0.7) +
  scale_colour_manual(values = c("Action A" = okabe_ito[1],
                                 "Action B" = okabe_ito[2])) +
  theme_publication() +
  labs(title = "Q-Value Convergence: Coordination Game",
       x = "Episode", y = "Q-value",
       colour = "Action", linetype = "Player")

p_conv
```

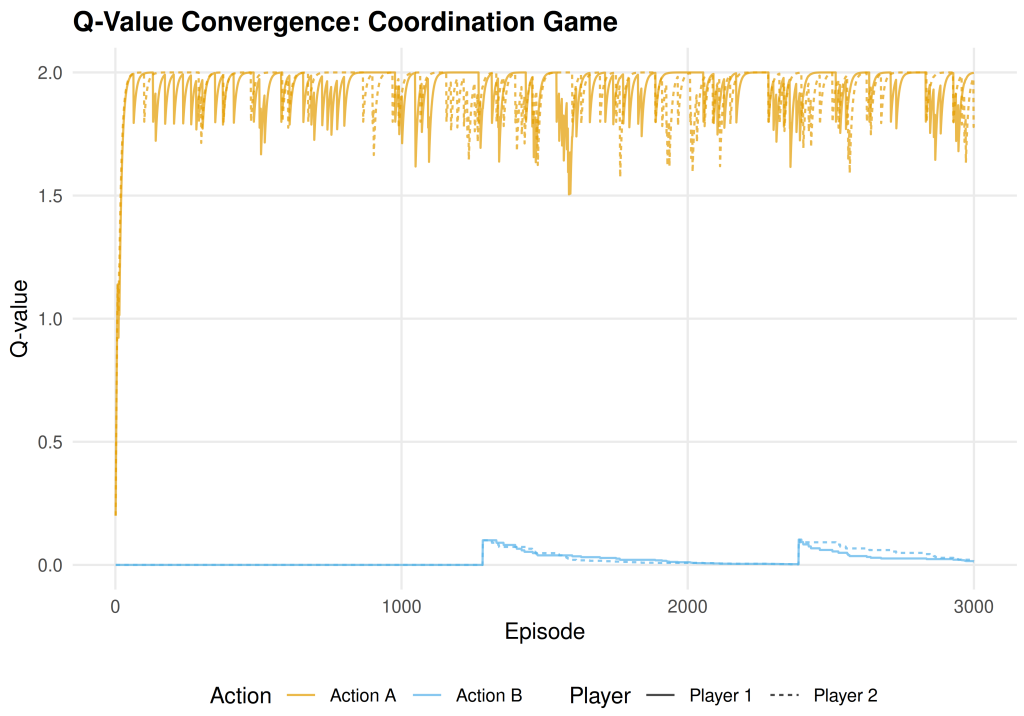


Figure 26.1: Q-value evolution in a coordination game. Both agents converge to action A (the payoff-dominant Nash equilibrium), with Q-values stabilizing near the equilibrium payoff of 2.

```
save_pub_fig(p_conv, "q-convergence-coordination", width = 7, height = 5)
```

26.3.4 Zero-sum game: cycling

```
# Matching Pennies
A_mp ← matrix(c(1, -1, -1, 1), nrow = 2, byrow = TRUE,
              dimnames = list(c("H", "T"), c("H", "T")))
B_mp ← -A_mp

set.seed(42)
hist_mp ← run_q_learning_game(A_mp, B_mp,
                              episodes = 3000, alpha = 0.05, epsilon = 0.1)
```

```
q_long_mp ← hist_mp ▷
  select(episode, Q1_1, Q1_2, Q2_1, Q2_2) ▷
  pivot_longer(-episode, names_to = "variable", values_to = "Q") ▷
  mutate(
    player = if_else(str_starts(variable, "Q1"), "Player 1", "Player 2"),
    action = if_else(str_ends(variable, "_1"), "Heads", "Tails")
  )

p_cycle ← ggplot(q_long_mp, aes(x = episode, y = Q, colour = action, linetype =
player)) +
  geom_line(alpha = 0.7) +
  scale_colour_manual(values = c("Heads" = okabe_ito[1],
                                 "Tails" = okabe_ito[2])) +
  theme_publication() +
  labs(title = "Q-Value Cycling: Matching Pennies",
       x = "Episode", y = "Q-value",
       colour = "Action", linetype = "Player")

p_cycle
```

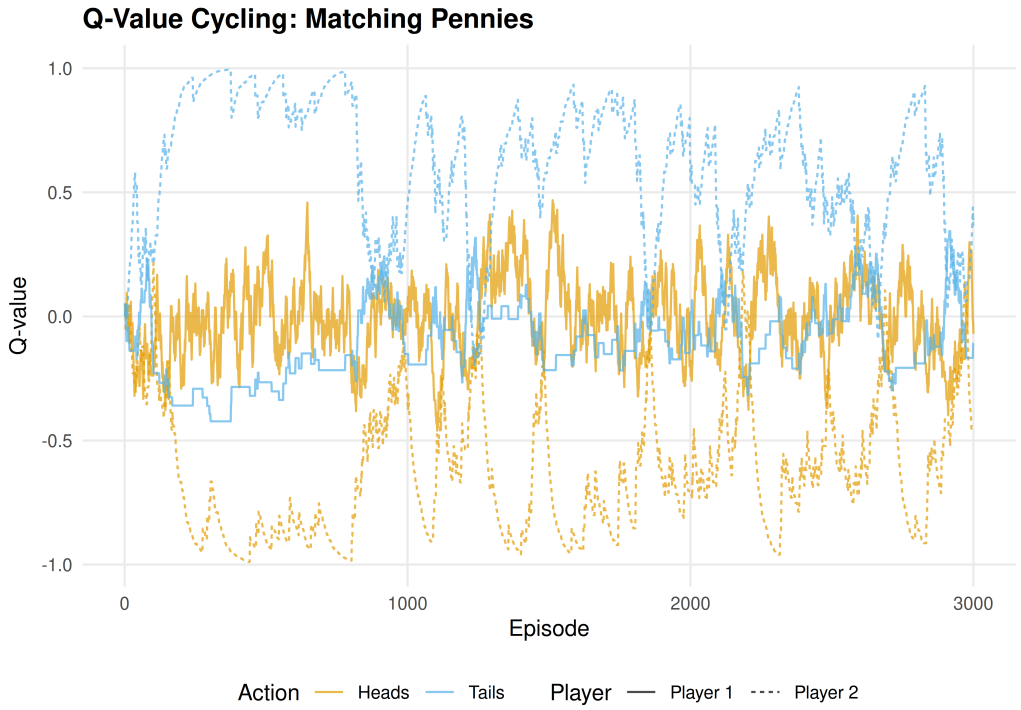


Figure 26.2: Q-value evolution in Matching Pennies (zero-sum). Instead of converging, Q-values oscillate as each agent's adaptation shifts the other's optimal play. The mixed NE at (0.5, 0.5) is never stably reached.

```
save_pub_fig(p_cycle, "q-cycling-matching-pennies", width = 7, height = 5)
```

26.4 Worked example

Let us trace the first 20 episodes of the coordination game in detail:

```
trace ← hist_coord ▷
  filter(episode ⩽ 20) ▷
  mutate(
    a1_label = if_else(a1 == 1, "A", "B"),
    a2_label = if_else(a2 == 1, "A", "B"),
    coordinated = a1 == a2
  )

cat("First 20 episodes of coordination game Q-learning:\n\n")
```

```
#> First 20 episodes of coordination game Q-learning:
```

```
trace ▷
  select(episode, a1_label, a2_label, r1, r2, coordinated) ▷
  rename(P1 = a1_label, P2 = a2_label, `P1 payoff` = r1,
         `P2 payoff` = r2, Coordinated = coordinated) ▷
  print(n = 20)
```

```
#> # A tibble: 20 × 6
#>    episode P1    P2    `P1 payoff` `P2 payoff` Coordinated
#>      <int> <chr> <chr>       <dbl>       <dbl> <lgl>
#>  1       1 A     A               2           2 TRUE
#>  2       2 A     A               2           2 TRUE
#>  3       3 A     A               2           2 TRUE
#>  4       4 A     A               2           2 TRUE
#>  5       5 A     A               2           2 TRUE
#>  6       6 A     A               2           2 TRUE
#>  7       7 A     A               2           2 TRUE
#>  8       8 A     A               2           2 TRUE
#>  9       9 A     B               0           0 FALSE
#> 10      10 B     A               0           0 FALSE
#> 11      11 A     B               0           0 FALSE
#> 12      12 A     A               2           2 TRUE
#> 13      13 A     A               2           2 TRUE
#> 14      14 A     B               0           0 FALSE
#> 15      15 A     A               2           2 TRUE
#> 16      16 A     A               2           2 TRUE
#> 17      17 A     A               2           2 TRUE
#> 18      18 A     A               2           2 TRUE
#> 19      19 A     A               2           2 TRUE
#> 20      20 A     A               2           2 TRUE
```

```
cat(sprintf("\nCoordination rate (first 20): %.0f%%\n",
            100 * mean(trace$coordinated)))
```

```
#>
#> Coordination rate (first 20): 80%
```

```
cat(sprintf("Coordination rate (last 500): %.0f%%\n",
            100 * mean((hist_coord |> filter(episode > 2500))$a1 ==
                       (hist_coord |> filter(episode > 2500))$a2)))
```

```
#> Coordination rate (last 500): 94%
```

Early episodes show exploration: agents try both actions and sometimes miscoordinate. As Q-values diverge (one action becomes clearly better), the ε-greedy policy increasingly selects the high-Q action, and coordination improves. By the final 500 episodes, coordination is near-perfect.

In Matching Pennies, by contrast, no such convergence occurs — the Q-values never stabilize because each agent's adaptation continually shifts the landscape the other agent faces.

26.5 Extensions

- **Decaying exploration**: Using $\varepsilon_t = \varepsilon_0 / t$ guarantees convergence in stationary environments but can lock in suboptimal play in non-stationary multi-agent settings.

- **Win-or-Learn-Fast (WoLF)**: Varies the learning rate based on whether the agent is winning or losing, achieving convergence in a broader class of games.
- **Multi-agent RL** (27) covers advanced approaches: MADDPG, PSRO, and the challenge of non-stationarity at scale.
- **Reinforcement learning foundations** (25) provides the single-agent background that this chapter builds on.

For the theoretical debate on rational learning in games, see Shoham & Leyton-Brown (2009) (Chapter 7).

Exercises

1. **Prisoner's Dilemma.** Run the Q-learning simulation on the standard Prisoner's Dilemma ($T = 5$, $R = 3$, $P = 1$, $S = 0$). Do both agents converge to mutual defection? How many episodes does it take? Plot the Q-value evolution.

2. **Learning rate sensitivity.** Run the coordination game with $\alpha \in \{0.01, 0.1, 0.5\}$ and fixed $\varepsilon = 0.05$. How does the learning rate affect convergence speed and stability? Plot Q-value trajectories for all three.

3. **Boltzmann exploration.** Replace ε-greedy action selection with Boltzmann (softmax) exploration: $P(a_i) = \exp(Q(a_i)/\tau) / \sum_{a'} \exp(Q(a')/\tau)$, where τ is a temperature parameter. Implement this and compare performance in the coordination game with ε-greedy.

Solutions appear in D.